# When Trading Dries Up

## A liquidity signal for a long-only fund

Silver Fund Research

Built on 2011–2018, tested on held-back 2019–2024

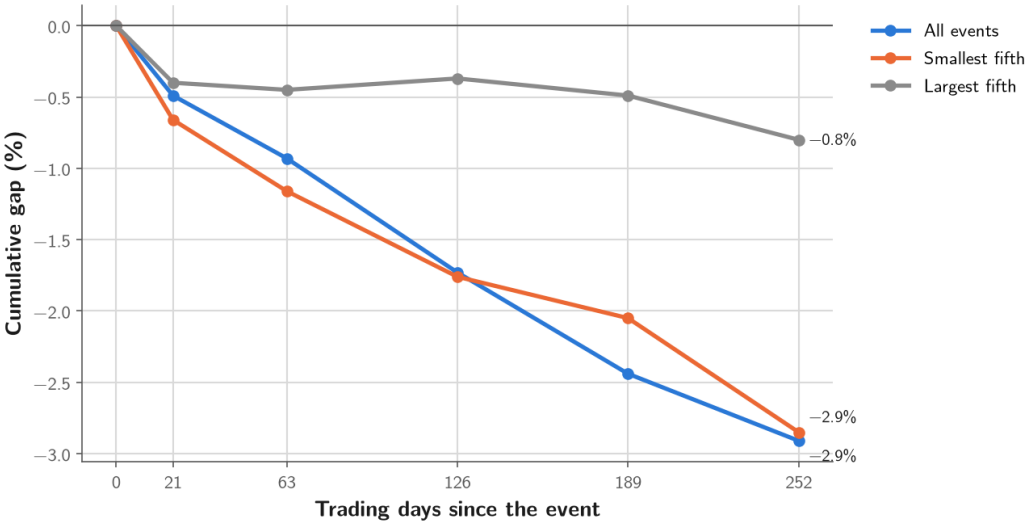
# The idea

- A **liquidity event**: a stock suddenly trades far less than usual for it (measured against its own past year).
- **Why it should matter:** most investors avoid illiquid stocks; only a small group of patient investors holds them, at a slightly lower price.
- When a stock's trading collapses, it has to pass from its current holders to that small group.
  - Sellers can only exit slowly, because volume has dried up; the few buyers need a lower price.
  - So the price **walks down for months**. Shorting these stocks is costly, so nobody trades the drift away.
- We first remove anything the fund's six existing scores already capture.
- Questions: is the effect real, and can our long-only fund make money from it?

# Bottom line

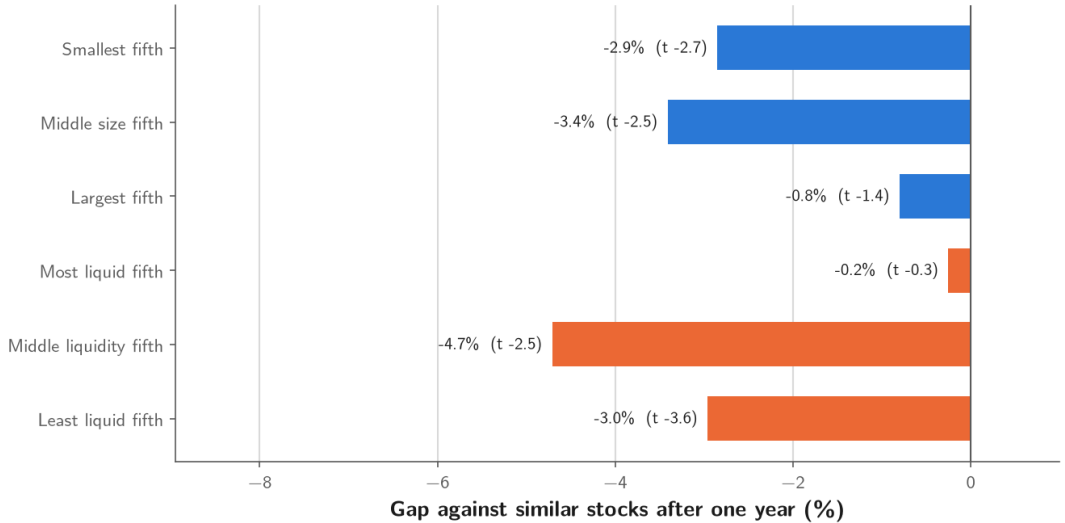
- **Yes, it is real.** Stocks whose trading dries up keep falling: about 3% behind similar stocks over the next year.
- **Just avoiding them doesn't help:** +0.05% to +0.10% a year, not significant.
- **Betting against them daily with the optimizer works**, also on 2019–24 data we held back: long-only +0.51% a year (IR 1.05).
- **It is small:** about +0.31% a year after trading costs. Use it as one input to the fund's alpha.

# After trading dries up, the stock keeps falling




Return after a liquidity event minus similar stocks (same size and liquidity) on the same day; 2006–2018. All events: t = -3.6 at one year.

# Mostly in small, illiquid stocks




Gap against similar stocks one year after the event, by size (blue) and by how illiquid the stock was before (orange); 2006–2018.

# It is not something we already know

| | How much less flagged stocks earn, %/month ($t$) |
|---|---|
| Had a liquidity event in the last month | 0.18 (2.5) |
| ... also controlling for last month's return | 0.17 (2.4) |

Positive = flagged stocks earn less, which is what we bet on. Monthly regressions of next month's return, 2008–2018; every row also controls for size, how illiquid the stock already was, its past-year return and its return two to three years ago, and the fund's six scores were removed before flagging. Measured once a month, so it understates the first-weeks effect that daily trading captures.

# The simple idea doesn't work: just avoid these stocks

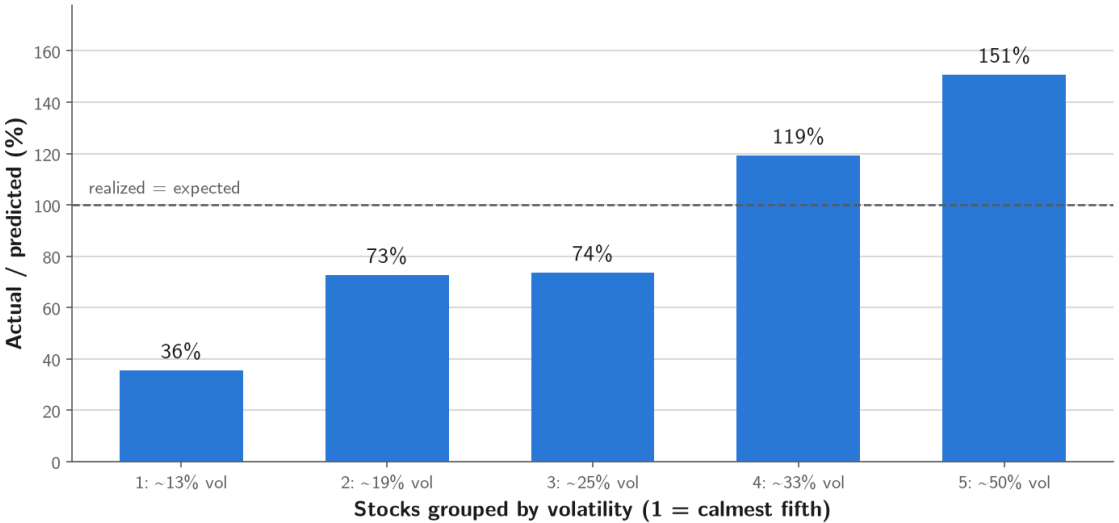
| 2011–2018 | Weight avoided | Return | Risk | IR ($t$) | IR, market removed |
|---|---|---|---|---|---|
| Just avoid flagged stocks, daily | 8.6% | +0.05%/yr | 0.47% | 0.11 (0.3) | −0.12 |
| Just avoid flagged stocks, monthly | 8.7% | +0.10%/yr | 0.43% | 0.23 (0.7) | −0.12 |
| Optimizer, daily, no shorting | — | +0.19%/yr | 0.10% | 1.82 (5.1) | — |

- Hold the benchmark, sell every stock with a liquidity event in the last 63 days, give its weight to similar stocks.
- It barely earns anything: these stocks are small, so dropping them tilts the portfolio in ways that add noise and swamp the effect.
- The optimizer removes that noise and sizes each bet. That is what makes the signal pay.

# How we bet on it

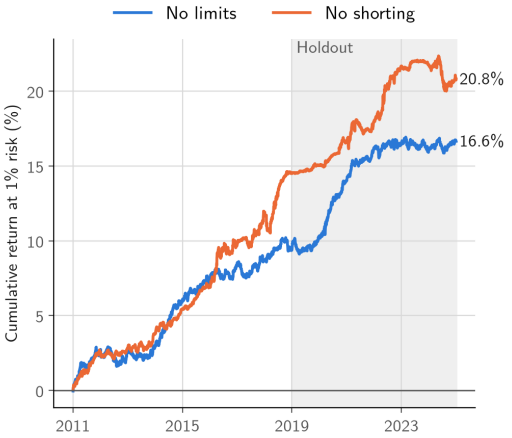
- **Alpha** (Grinold–Kahn, Manual 4.6): $\alpha_i = IC \times \sigma_i \times Z_i$.
  - Our $Z_i$ is a flag: did stock $i$ have a liquidity event 1–21 days ago, or 22–63 days ago?
  - We estimate $IC \times Z$ directly: the average gap to similar stocks per unit of volatility, for each age (a regression on a 0/1 flag, Manual 4.4). Past years only, re-estimated every January.
- **Optimizer** (Manual ch. 3): maximize $h^\top \alpha - \frac{\gamma}{2} h^\top V h$ with $h^\top \iota = 0$, $h^\top \beta = 0$, 5% tracking error, and size and liquidity in line with the benchmark.
- **Two versions:** no limits, and no shorting ($b + h \geq 0$, Manual 3.4). The transfer coefficient extends the fundamental law to $IR = TC \times IC\sqrt{BR}$: TC is how much of the forecast survives the long-only rule.
- **No trading costs:** the signal would be one input to the fund's alpha, where trades net out. We show what it would cost alone.

# Calm stocks fell less than we expected, volatile stocks more




An earlier version left volatility out of the Grinold–Kahn alpha. Bars: actual return as % of predicted, 2011–2018. So alpha now scales with volatility.

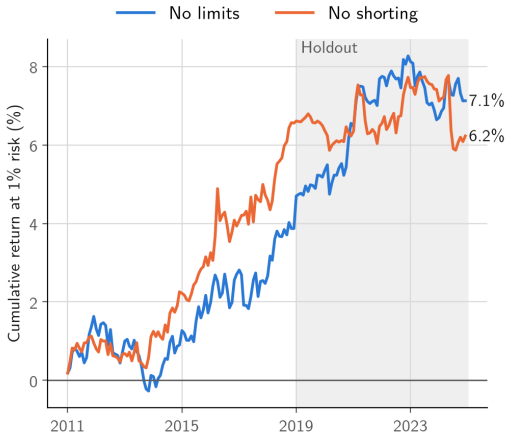
# Daily rebalancing, 2011–2018 and holdout 2019–2024




| | No limits | | No shorting | |
|---|---|---|---|---|
| | 2011–18 | 2019–24 | 2011–18 | 2019–24 |
| Active return, %/yr | +6.84 | +7.43 | +0.19 | +0.51 |
| Active risk (realized) | 5.82% | 6.12% | 0.10% | 0.48% |
| IR ($t$) | 1.17 (3.3) | 1.21 (3.0) | 1.82 (5.1) | 1.05 (2.6) |
| Max drawdown | −7.62% | −6.53% | −0.15% | −1.14% |
| Positive years | 7/8 | 5/6 | 8/8 | 5/6 |
| Turnover, ×/yr | 186.4 | 166.9 | 2.2 | 4.5 |
| TC vs no limits | ref. | ref. | 0.44 | 0.36 |
| Spread cost if traded alone, %/yr | 14.06 | 15.91 | 0.09 | 0.19 |

Each period's line scaled to 1% realized risk; the table shows actual numbers. No shorting = total weight $\geq 0$. TC = ex-ante transfer coefficient vs. no limits. Holdout: design frozen before it was run; settings re-estimated each January from earlier years only.

# Monthly rebalancing, 2011–2018 and holdout 2019–2024




| | No limits | | No shorting | |
|---|---|---|---|---|
| | 2011–18 | 2019–24 | 2011–18 | 2019–24 |
| Active return, %/yr | +2.92 | +3.58 | +0.08 | −0.02 |
| Active risk (realized) | 6.03% | 6.59% | 0.10% | 0.44% |
| IR ($t$) | 0.48 (1.4) | 0.54 (1.3) | 0.82 (2.3) | −0.05 (−0.1) |
| Max drawdown | −11.47% | −10.74% | −0.14% | −0.84% |
| Positive years | 5/8 | 5/6 | 7/8 | 2/6 |
| Turnover, ×/yr | 60.9 | 53.9 | 0.7 | 1.5 |
| TC vs no limits | ref. | ref. | 0.44 | 0.36 |
| Spread cost if traded alone, %/yr | 5.63 | 6.28 | 0.02 | 0.05 |

Each period's line scaled to 1% realized risk; the table shows actual numbers. No shorting = total weight $\geq 0$. TC = ex-ante transfer coefficient vs. no limits. Holdout: design frozen before it was run; settings re-estimated each January from earlier years only.

# How much to trust it

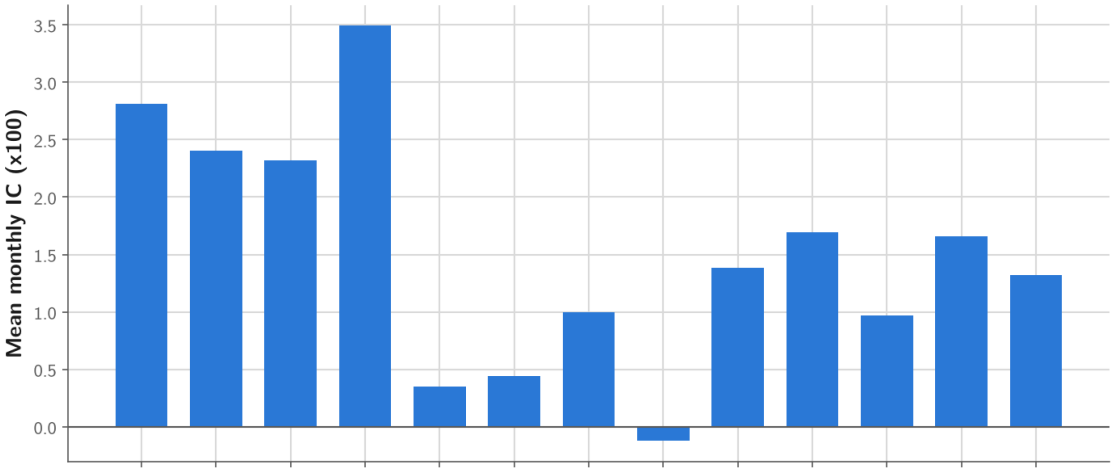
- We built it on 2011–18 in four rounds, then froze it and ran 2019–24 **once**.
- In 2019–24 the long-only portfolio is bigger (0.48% risk vs 0.10%): the model learned that large stocks drift too, so it had more weight to sell (appendix).
- After trading costs, long-only earns about $+0.31\%$ a year (IR about 0.6).
- Six years is short (standard error of an IR $\approx 1/\sqrt{\text{years}}$): the true IR could be anywhere from about 0.2 to 1.9.
- Not yet checked: thinly traded stocks have stale prices, which can make daily risk look too small.

# Conclusion

- Stocks whose trading dries up keep falling for months: they must pass to a small group of patient buyers, and that hand-off is slow.
- A long-only fund can make money from it, but only with the optimizer and daily trading.
- It is small. Use it as one input to the fund's alpha, not on its own.
- Next: combine it with the fund's scores and test the combined alpha, with trading costs.

# Appendix

# Appendix: new information beyond the fund's scores? IC by year




Rank IC with next-month Barra specific return, fund's six scores removed. Mean 0.015 (t 6.0); 12 of 13 years positive.

# Appendix: what did we change, and when?

| Daily runs, %/yr (IR), no costs | No limits | No shorting |
|---|---|---|
| 1. Flag that turns over daily | −0.9 (−0.15) | — |
| 2. Event clock | −2.77 (−0.49) | −0.12 (−0.24) |
| 3. + no positive alpha, hedge with similar stocks | +3.23 (0.58) | +0.36 (1.48) |
| 4. + alpha scaled by volatility | +6.84 (1.17) | +0.19 (1.82) |

- Version 2: a flag that switched off after a few days missed most of the drift, so we gave each event an age.
- Version 3: a noisy positive alpha for large stocks made the portfolio buy them; we stopped allowing positive alpha and hedged with similar stocks.
- Version 4: calm stocks fell less than predicted (slide 9), so alpha now scales with volatility. Frozen before the 2019–24 test.

# Appendix: why is the long-only result bigger in 2019–24?

| Year | Large-stock alpha | Long-only risk | Long-only return |
|---|---|---|---|
| 2011 | 0.0 | 0.08% | +0.25% |
| 2012 | 0.0 | 0.06% | +0.04% |
| 2013 | 0.0 | 0.08% | +0.10% |
| 2014 | −0.1 | 0.07% | +0.17% |
| 2015 | 0.0 | 0.05% | +0.16% |
| 2016 | −0.8 | 0.15% | +0.28% |
| 2017 | −2.1 | 0.12% | +0.21% |
| 2018 | −3.2 | 0.16% | +0.29% |
| *Holdout* | | | |
| 2019 | −4.9 | 0.16% | +0.26% |
| 2020 | −5.0 | 0.30% | +0.76% |
| 2021 | −7.2 | 0.59% | +0.63% |
| 2022 | −7.1 | 0.68% | +1.80% |
| 2023 | −7.3 | 0.21% | +0.06% |
| 2024 | −7.4 | 0.66% | −0.47% |

- Until 2015 the model gave large stocks no alpha, so it could only sell small stocks, which hold little benchmark weight.
- As evidence built up, large stocks got a negative alpha too, so the portfolio had more weight to sell.
- Same signal, bigger portfolio.

Large-stock alpha: predicted daily drift per unit of volatility, $\times 10{,}000$, for the first month after an event. Long-only: daily, no shorting.

# Appendix: where does the return come from? (2011–2018)

| Return, %/yr | Event 1–21 days ago | Event 22–504 days ago | No recent event |
|---|---|---|---|
| Daily, no limits | +14.88 | −3.43 | −4.64 |
| Daily, no shorting | +0.12 | +0.08 | −0.01 |
| Monthly, no limits | +8.02 | +0.09 | −4.09 |
| Monthly, no shorting | +0.09 | +0.06 | −0.05 |

Return earned on positions in stocks with a recent event, an older event, or none. In the no-limits runs the bets against flagged stocks are hedged with similar stocks; small, illiquid stocks lag in general, so both sides show it and it cancels.